

Extended Abstract

Motivation Large language models (LLMs) recently showed great potentials, but often are limited by the availability of large-scale training data. Self-play finetuning, that is, to finetune the models using responses generated themselves, offers solution to this problem. However, this approach is constrained by the lack of diversity in responses and hence the quick plateaus in performance improvement. One recent research (Subramaniam et al., 2025) proposes that using multiple LLM agents, instead of one, in generating self-training data can increase the diversity in responses and bring more improvement. Nevertheless, the differentiation of agents in their approach still emerged in an unstructured and uncoordinated way. To improve, this paper draws insights from human organizations and proposes a new method that introduces coordination among agents, by implementing a coordinator agent to enhance the specialization of agents into divers aspects of the problem and thus retaining more diversity in responses.

Method First, the coordinator agent will decompose the problem into N aspects. Next, the original problem and the decomposition will be passed to each sub-agent. After the first round, responses from all sub-models will be summarized and passed to each sub-model for the next round of reasoning. The multiple rounds design aims for enhancing the deliberation thinking. After M rounds, the coordinator agent will aggregate final outputs from all sub-models and produce the collective answer. Therefore, the coordinator agent only focuses on the "macro" level tasks (decomposing and aggregating) and the sub-models only focus on one of the problem aspects. This design allows a division of labor between "seeing the forest" and "seeing the tree(s)" and should help improving the coordination among agents. After collecting a batch of responses, the correct outputs will be used for finetuning (assuming there is an object reward evaluator). More specifically, if the final output is correct, then all the corresponding intermediate responses from sub-model i (the coordinator model) will be used to finetune sub-model i (the coordinator model) exclusively. That is, each model will be finetuned on a separate set of data, which is designed to help them specialize.

Implementation I evaluate the method on a math reasoning task: Countdown (Gandhi et al., 2024). This problem provides the model with a set of numbers and a target. The model need to generate a mathematical expression that combines the given numbers with operations and equals the target. I use the TinyZero dataset (Pan, 2025) to generate responses for finetuning, by randomly selecting 200 examples. I planned to compare the proposed method with the original multiagent finetuning as the main baseline. However, given the unexpected poor performance of small models on multiagent interaction, I compare with the Supervised Fine-Tuning (SFT) model.

Results The proposed coordinated reasoning approach almost completely destroyed the model's capacity, with the score dropping from 0.289 (in SFT) to 0.0605. This poor performance prevents me from collecting correct responses for finetuning. Further analyses show that small models fail to coordinate because of the lack of proper general capacities in problem decomposing, instruction following, and responses aggregating.

Discussion Taken together, the results and analyses demonstrate that coordination is a high level capacity that is challenging for small sized models. To successfully harness the power of coordination in more systematically exploring the problem space and retain a higher level of diversity in responses for effective finetuning, the base model needs at least three general capacities: (1) understanding and following instructions, (2) understanding the problem (its structure and different components), and (3) understanding other models' responses and summarizing (aggregating) them.

Conclusion This paper proposes a novel method of multiagent reasoning and finetuning. It introduces a coordinator agent to decompose the problem into different aspects for other sub-agents to work on, and then to aggregate their responses into the final answer. This method aims at helping agents specialize and retain diversity in their responses for better finetuning. However, experiments show that this method is not beneficial for small sized models, constrained by their limited general capacity. Future research should explore the use of coordination in reasoning and finetuning in larger and more capable models.

Coordinated Multiagent Reasoning and Finetuning

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Abstract

Large language models (LLMs) recently showed great potentials, but often are limited by the availability of large-scale training data. Self-play finetuning, that is, to finetune the models using responses generated themselves, offers solution to this problem. However, this approach is constrained by the lack of diversity in responses and hence the quick plateaus in performance improvement. To improve this approach, this paper proposes a new method of multiagent finetuning, by introducing a coordinator agent to enhance the specialization of agents into diverse aspects of the problem and thus retaining the diversity in responses. Unexpectedly, initial experiment results reveal the limitation of this method: small models lack proper general capacities of problem decomposing, instruction following, and responses aggregating, and thus fail in coordination. Future research should explore the use of coordination in more capable base models and more diverse tasks.

1 Introduction

Large language models (LLMs) recently demonstrated extraordinary capacities on many tasks. However, their further improvement is often limited by the availability of data at large scale. One approach toward this problem is to generate synthetic data using LLM themselves for further training. A major problem of this self-play finetuning is the lack of diversity in synthetic data: they are often repetitive and similar in patterns. Thus, it often encounters plateaus in performance improvement quickly.

To improve this approach, one recent research (Subramaniam et al., 2025) proposes that using multiple LLM agents, instead of one, in generating self-training data can increase the diversity in their responses, which turns out to be beneficial in delaying the plateaus and producing more improvement. More specifically, they start with several pairs of generator and critic agents from the same base model, which generate responses collectively through multiple rounds of generating-summarizing-criticizing interactions and eventually a majority vote. Crucially, each pair of generator and critic agents was finetuned independently, only using synthetic data they produced. By doing so, agents spontaneously differentiate from each other and thus maintained a higher level of diversity in responses.

However, despite considerable improvement, this method still hasn't unleashed the full potential of self-play finetuning. The key is that the differentiation of agents emerged in an unstructured and uncoordinated way: the agents are merely "differentiated" rather than "specialized". They still individually handle the whole problem rather than specialize in certain components or aspects of the problem. Thus, some of the efforts are wasted in generating and resolving opposing views on the same aspect of a problem. This could constrain and limit the search in the problem space, preventing the model from systematically discovering and generating more diverse information that is truly critical to the sustainable improvement through self-play finetuning.

In this paper, I propose a new approach of self-play finetuning that has the potential of further increasing the diversity of responses and hence more improvements in performance: by introducing

coordination among the agents. I achieve this by implementing a new role of agents: coordinator. Inspired by coordinators in human organizations, the coordinator agent will divide the task into subtasks that other agents can subsequently specialize in, and then aggregate specialized agents' outputs into the final response.

However, as I will demonstrate below, this approach turns out to be highly contingent on the general capacity of the base model. Coordination is a high-level capacity that builds on text understanding, instruction following, summarizing, etc. and combining them with reasoning. In the small 0.5B model I experiment with, the lack of such capacities prevents the use of coordination and makes this approach even detrimental to math reasoning performances, to the extent that finetuning is hardly possible on generated responses. Thus, further research should explore the use of coordination in more capable LLMs and investigate the interaction between base capacity and effectiveness of self-play finetuning.

2 Related Work

In cognitive science and organization theory, coordination is a well-investigated topic (Malone and Crowston, 1994; March and Simon, 1958). Sufficient evidence suggests the advantages of coordinated agents over crowds (Woolley et al., 2010; Salas et al., 2005). Research also illustrated the mechanisms of such advantages (Malone and Crowston, 1994; Heath and Staudenmayer, 2000), many of which are transferrable to the design of coordination between LLM agents. For example, when exploring in a high-dimensional complex landscape, appropriate coordination mechanisms can enhance the scope of exploration and prevent agents from getting stuck at local optima (Levinthal, 1997; Fang et al., 2010). Research also finds agents tend to "miss the forest for the trees", that they only conduct local modular search to optimize one subtask, while overlooking systematic searches that acknowledge the complex interdependences between subtasks and the integration between them (March and Simon, 1958; Gavetti and Levinthal, 2000; Heath and Staudenmayer, 2000; Clement, 2023). Addressing these problems, people need the assistance of miscellaneous coordination mechanisms, among which adding a coordinator role is especially simple and effective (Berntzen et al., 2023; Okhuysen and Bechky, 2009).

Often inspired by this literature, researchers have explored the coordination potential between LLM agents, which is critical for multiagent collaboration and human AI interaction (Li et al., 2023; Wu et al., 2023; Wang et al., 2024). Existing work mainly focuses on two approaches. One is to improve the algorithm and architecture of individual agents to equip them with better capacities to react to others' actions and coordination. For example, people have studied multiple ways of incorporating the theory of mind (TOM) and shown its great potential (Rabinowitz et al., 2018). Another approach aims for better frameworks to coordinate and organize a multiagent system (Li et al., 2023; Wu et al., 2023; Hong et al., 2024). However, there isn't efforts trying to integrate coordination into self-play finetuning.

Another related branch of work examines the reasoning structures of LLM agents, such as Tree-of-Thought (ToT) (Yao et al., 2023) and Graph-of-Thought (GoT) (Besta et al., 2024). This approach similarly involves the decomposition of ideas, exploration of ideas along a subordinate aspect, and the aggregation of different lines of reasoning. Differently, the current proposal focuses on finetuning rather than the final inference process and specializing multiple agents for these processes to enhance the systematic level exploration.

3 Method

Building on the framework proposed by Subramaniam et al. (2025), I further introduced a coordinator agent into the framework. The method has two part: first, reasoning with coordinated multiple agents; second, using responses from part one to finetune the models.

Figure 1 illustrates the reasoning framework (the numbers of sub-models N and rounds M are given hyperparameters). First, the coordinator agent will decompose the problem into N aspects. Next, the original problem and the decomposition will be passed to each sub-agent. After the first round, responses from all sub-models will be summarized and passed to each sub-model for the next round of reasoning. The multiple rounds design aims for enhancing the deliberation thinking. After M rounds, the coordinator agent will aggregate final outputs from all sub-models and produce the collective

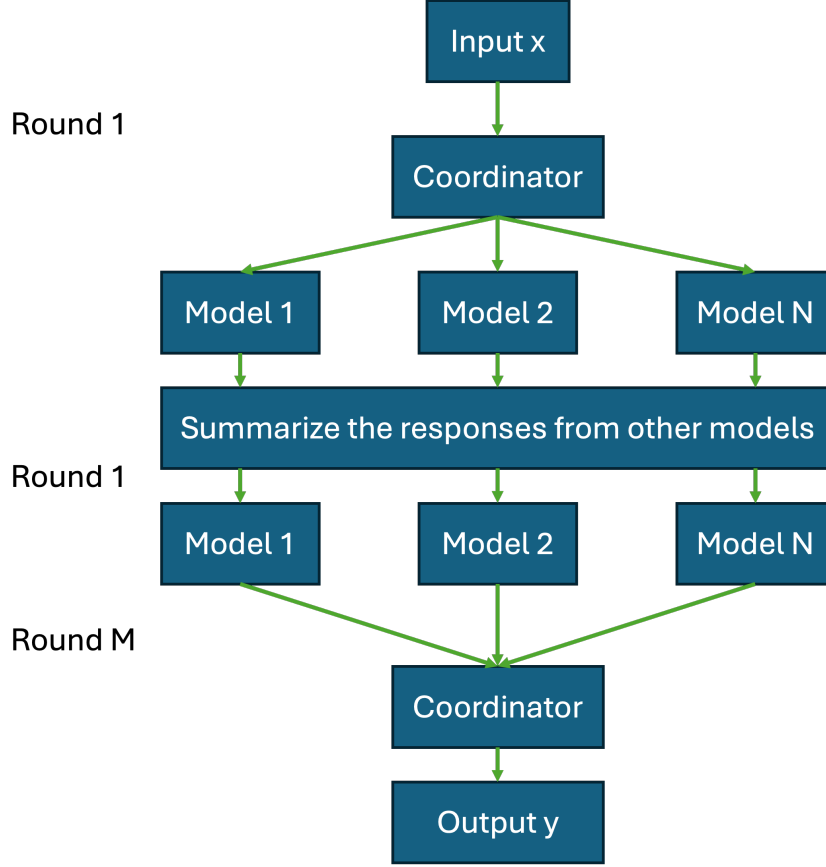


Figure 1: Method Overview.

answer. Therefore, the coordinator agent only focuses on the "macro" level tasks (decomposing and aggregating) and the sub-models only focus on one of the problem aspects. This design allows a division of labor between "seeing the forest" and "seeing the tree(s)" and should help improving the coordination among agents. Prompts for the coordinator and sub-agents can be found in the appendix.

After collecting a batch of responses, the correct outputs will be used for finetuning (assuming there is an object reward evaluator). More specifically, if the final output is correct, then all the corresponding intermediate responses from sub-model i (the coordinator model) will be used to finetune sub-model i (the coordinator model) exclusively. That is, each model will be finetuned on a separate set of data, which is designed to help them specialize.

4 Experimental Setup

I evaluate the method on a math reasoning task: Countdown (Gandhi et al., 2024). This problem provides the model with a set of numbers and a target. The model need to generate a mathematical expression that combines the given numbers with operations and equals the target. I use the TinyZero dataset (Pan, 2025) to generate responses for finetuning, by randomly selecting 200 examples.

I planned to compare the proposed method with the original multiagent finetuning as the main baseline. However, given the unexpected poor performance of small models on multiagent interaction, I compare with the Supervised Fine-Tuning (SFT) model. The SFT model was trained the WarmStart Dataset (Gandhi et al., 2025), with two epochs and a batch size of four.

I run experiments with the Qwen 2.5 0.5B Base model after SFT, on a NVIDIA Ampere A40 GPU, with $N = 2$ and $M = 2$.

5 Results

Ideally, the evaluation should have two parts: first, examining the performance on reasoning and secondly, performance after finetuning using the reasoning responses. However, as detailed below, the reasoning performance is too low to produce any correct responses for finetuning. Thus, I focus on the reasoning evaluation and the post-hoc analysis of the method.

5.1 Reasoning Performance

As shown in Table 1, the proposed coordinated reasoning approach almost completely destroyed the model’s capacity. The final output answers can hardly match the correct format, let alone correct results. I fail to extract any feasible responses for finetuning from the 200 samples.

Table 1: Performance Comparison

Method	Score
My Approach	0.0605
SFT	0.2890

5.2 Problem Analyses

Table 2: Performances of Different Decomposers (Row) and Aggregators (Column)

	Base	SFT
SFT	0.0100	0.0605
Base	0.0130	0.0690
Given Prompt	0.0175	0.0755

To further understand this massive performance drop, I look into responses at each stage and find three main problems.

First, the coordinator agent has difficulty in generating meaningful task decompositions. Second, all agents are not good at instruction following. Below is an example:

"1. First, let's try to get close to 19 by looking at the larger numbers:3. 38 is too large, we need to reduce it4. 32 is close to 19, could be useful5. 14 is close too<Repetitive reasoning texts omitted> Got it!38 - 11 = 2732 + 14 = 4627 - 46 - 19</think><answer> (38 - 11) - (32 + 14) </answer>"

The model starts with trying to generate high-level aspects, but soon turns to directly solve the problem. All SFT models output similarly patterned responses, no matter what additional instructions given or what roles attached. This could be the results of two factors: first, the SFT model is already highly specialized for the math task; second, the 0.5B model is too small for these requirements. To separate these two reasons, I experiment using the base 0.5B Qwen model for decomposing the task. As shown in the second row of Table 2, it indeed improves the performance, but only marginally. As illustrated by the example below, small models’ general instruction following and task decomposing capacities are overall weak, even before the SFT.

*"Certainly! Let's break down the given problem step by step into two distinct aspects:# Aspect 1: Using the Numbers [34, 98, 27, 7]need to create an equation that equals 30 using the numbers 34, 98, 27, and 7, and each number can only be used once.# Aspect 2: Solving the Equationwe have the equation, we need to solve it to find the value that makes the equation true.# Aspect 1: Creating the Equationcreate the equation, we need to find a combination of numbers that sum up to 30. Let's start by trying different combinations:1. **Summing up to 30:**- We need to find a combination of three numbers that add up to 30.- Let's try different combinations:- 34 + 98 + 27 = 30- 34 + 98 + 7 = 139 (not 30)<Repetitive reasoning texts omitted>"*

Given the structured nature of the math reasoning task, I also experiment replacing the decomposer with a fixed prompt (see below) generated by Gemini 2.5 Pro, a much more capable model. As shown

by the third row of Table 2, the performance increases only a little, indicating the existence of other problems.

"Aspect 1: Generating all possible permutations of the numbers and combinations of operations. Aspect 2: Evaluating each generated mathematical expression to check if it equals the target."

The third problem is that models here are incapable of correctly aggregating and summarizing responses to get the answer. For example, in the example below, even sub-agents have identified the correct answer, the coordinator fails to output it, instead verifying a non-existent solution.

"Let's verify: 1. $50 - 2 = 482$. $78 - 1 = 773$. $77 - 50 = 274$. $27 + 1 = 285$. $28 + 1 = 291$ apologize, but I've tried multiple systematic approaches and couldn't reach 51. There would be too small numbers or multiple results."

Similarly, I experiment using the base model for the final aggregation. As shown in the first column of Table 2, conversely, the performance even dropped more. This suggests that the understanding of the math problem itself is more critical, which is gained through SFT.

6 Discussion

Taken together, the results and analyses demonstrate that coordination is a high level capacity that is challenging for small sized models. To successfully harness the power of coordination in more systematically exploring the problem space and retain a higher level of diversity in responses for effective finetuning, the base model needs at least three general capacities: (1) understanding and following instructions, (2) understanding the problem (its structure and different components), and (3) understanding other models' responses and summarizing (aggregating) them.

Sadly, the mini model experimented in this paper does not possess these key capacities. Due to limitation in computing power, I could not experiment this method on stronger base models. The task type may also contribute to the poor performance. Math reasoning is notoriously difficult for LLMs. It is also tough to correctly decompose a math problem or use decomposition to assist math reasoning given the tight and often single logic chain. To more extensively examine the coordination approach, a more comprehensive and representative set of tasks should be considered.

7 Conclusion

This paper proposes a novel method of multiagent reasoning and finetuning. It introduces a coordinator agent to decompose the problem into different aspects for other sub-agents to work on, and then to aggregate their responses into the final answer. This method aims at helping agents specialize and retain diversity in their responses for better finetuning. However, experiments show that this method is not beneficial for small sized models, constrained by their limited general capacity. Future research should explore the use of coordination in reasoning and finetuning in larger and more capable models.

8 Team Contributions

- **Yankai Wang:** All.

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A Implementation Details (Prompts)

Coordinator (Decomposer) Given the following math problem: {question}

Please break this problem down into {num_aspects} distinct key aspects or sub-problems that can be worked on by different agents. Each aspect should help in solving the overall problem. List the aspects clearly, each on a new line, like this: Aspect 1: [Description of aspect 1] Aspect 2: [Description of aspect 2] Do not provide any additional explanations or context, just the aspects. Only decompose the problem into aspects, do not solve it.

Sub-agents You are Agent $i+1$ of agents_count, collaborating to solve a complex math problem. The overall problem is: '{question}'. A coordinator has broken down the problem into several aspects: {decomposed_aspects} Your primary assigned aspect to focus on aspect $i+1$. Please provide a detailed, step-by-step reasoning process focusing on your assigned aspect. Then, attempt to provide a solution to the OVERALL problem based on your work on this aspect and how it integrates with the larger problem. Conclude your response with the final answer to the main problem in the format boxedanswer.

Coordinator (Aggregator) The original math problem was: 'question'. A coordinator initially broke the problem down into the following aspects for guidance: '; '.join(decomposed_aspects) agents_count agents then worked on this problem over rounds rounds. Here are their final responses: Please synthesize these agent responses, considering the original problem and the initial aspects. Provide a single, consolidated, and well-reasoned solution. Explain the steps clearly and derive the final answer in the format boxedanswer.